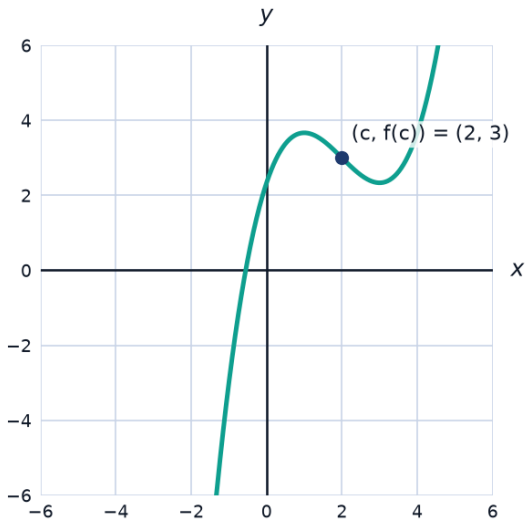
Euler's Method and Logistic Growth

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

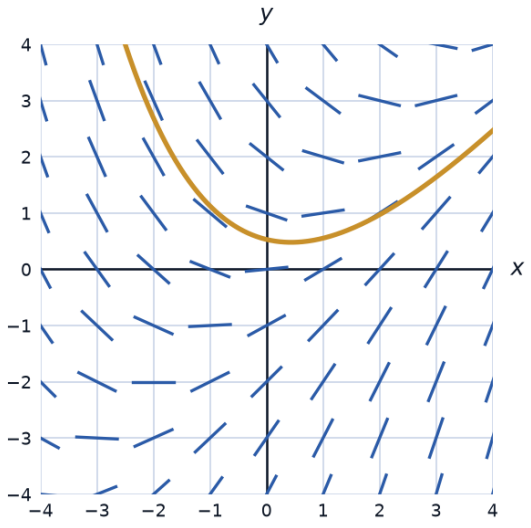
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = a$ and a local minimum at $x = b$. The tangent line to the graph of f at $x = c$ has the equation $y - f(c) = \frac{1}{2}(x - c)$. Given that $f(c) = 3$ and $f'(c) = \frac{1}{2}$, find the value of c.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

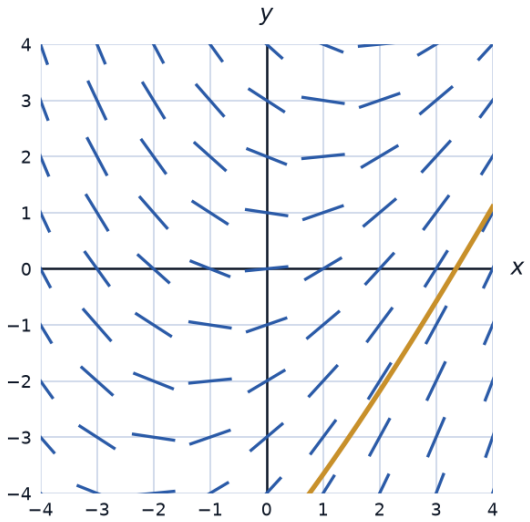
A particle's position in the xy-plane is given by the parametric equations $x(t) = at^2 - 3t$ and $y(t) = 2t^3 + bt$, where a and b are constants. At time $t = 1$, the particle's velocity vector is $\mathbf{v}(1) = \langle 9, 10 \rangle$. Find the value of a and the value of b.



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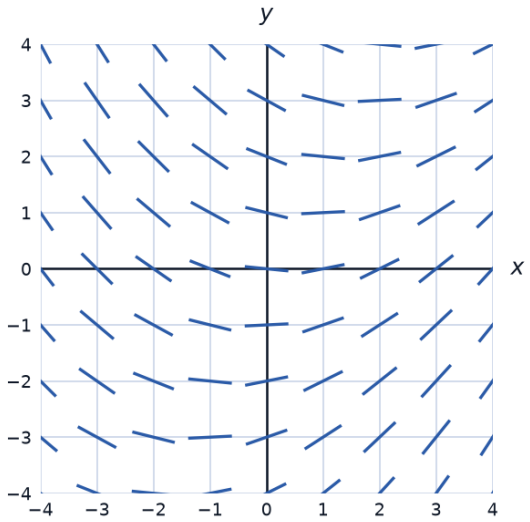
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Give the first three nonzero terms of the Maclaurin series for $\sin x$.



4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

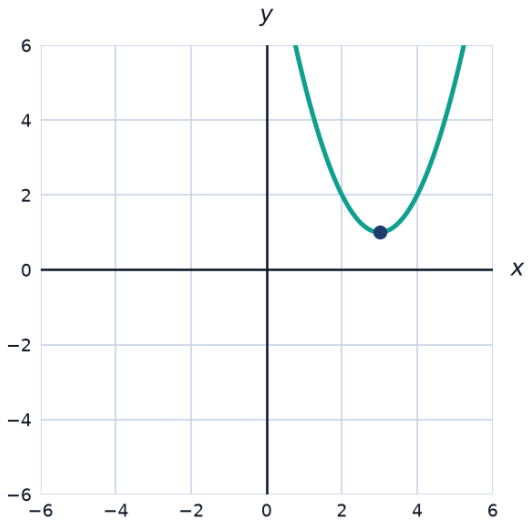
Give the first three terms of the Maclaurin series for $\cos x$.



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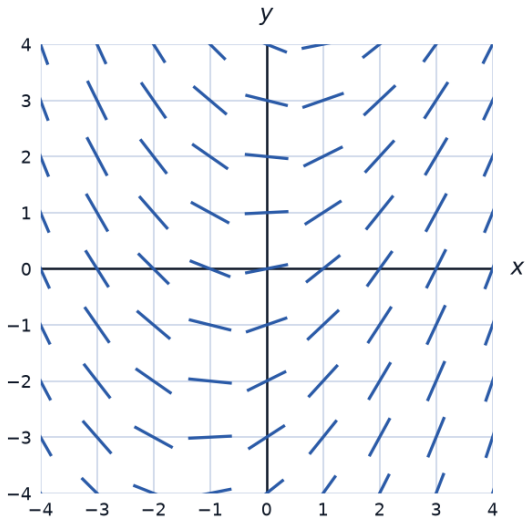
5 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x) = x^2 - 6x + 10$. Find the x-coordinates of the points on the graph of $y = f(x)$ where the tangent line is horizontal.



6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For what x does $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converge?



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The polar curve $r = 2\cos\theta$ is a circle. Find its enclosed area.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the area enclosed by the cardioid $r = 1 + \cos\theta$.

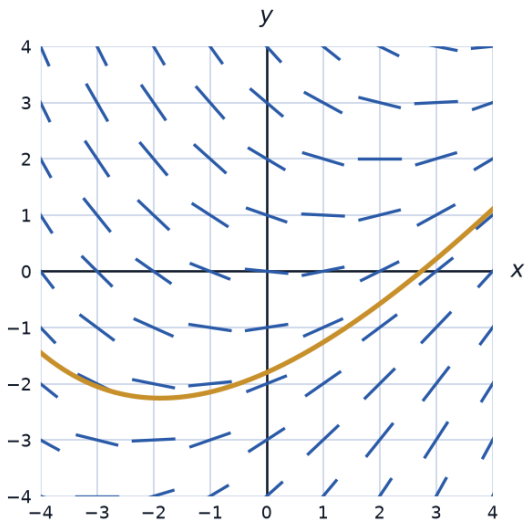
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9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x)$ defined by its Maclaurin series $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!\,3^{2n+1}}$. Let $g(x) = \int_0^x f(t)dt$. The function $f(x)$ is a known elementary function. Determine the value of $g''(0)$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 1 · FoundationsOn an AP Calculus BC exam question, a student writes $\int_0^5 (R(t) - D(t))\,dt = 42.18$ as their final response to a question asking for the total amount of water in a tank at $t = 5$, given initial volume 100 gallons. Identify the missing component in the student's solution.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · Advanced / AP FRQLet $y = f(x)$ be the particular solution to $\frac{dy}{dx} = xy^2$ with $f(1) = -1$. Use Euler's method with step size $\Delta x = 0.5$ starting at $x = 1$ to approximate $f(2)$, and determine whether the approximation is an overestimate or underestimate by computing $\frac{d^2y}{dx^2}$ at $(1, -1)$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · Advanced / AP FRQLet $f(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$. Identify the closed-form elementary function that this series represents, and determine its radius of convergence.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumOn Section II Part A (Calculator Active), a student is asked for the acceleration vector at $t = 3$ for a particle whose velocity is $\vec{v}(t) = \langle \ln(t^2 + 1), 3\cos(t) \rangle$. What calculator command must be used to find this vector, and what is its 3-decimal numerical value?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 2 · MediumLet $\frac{dy}{dx} = x + 2y$ with $y(0) = 1$. Use Euler's method with step size $\Delta x = 0.2$ to approximate $y(0.4)$.

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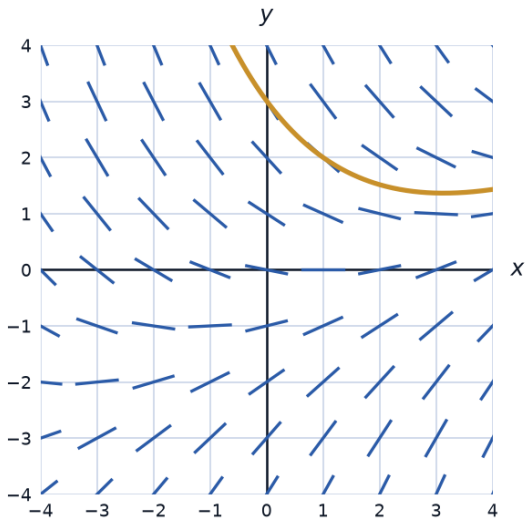
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

[AP Calculus BC Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Level 1 · FoundationsWhich single unit on the AP Calculus BC exam carries the highest individual weighting, and what specific topic within it is virtually guaranteed to appear on Section II as Free Response Question 6?



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Level 1 · FoundationsA population satisfies $\frac{dP}{dt} = 0.05P - 0.0001P^2$. Determine the carrying capacity L and evaluate $\lim_{t\to\infty} P(t)$ given initial condition $P(0) = 150$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 Taylor and Maclaurin Series

[Unit 7BC · Logistic Models & Euler's Method · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Integration by Parts

Level 2 · MediumDuring Section II Part A (30 minutes, Calculator Active), a student completes Questions 1 and 2 with 5 minutes remaining. When time is called and calculators are stored for Part B (60 minutes, No Calculator), may the student return to Questions 1 and 2 to add work, and what restrictions apply?

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Level 3 · AdvancedA differential equation $\frac{dy}{dx} = (y-1)^2\cos(x)$ has initial condition $y(0) = 2$. Find the particular solution $y = f(x)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $\frac{dy}{dx} = x + y$ and initial condition $y(0) = 1$, use Euler's method with step size $\Delta x = 0.5$ to approximate $y(1)$.

a. 2.25

b. 2.00

c. 2.50

d. 2.75

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $y'' > 0$ (concave up) on the interval, does Euler's method yield an overestimate or underestimate of the true solution?

a. An underestimate because tangent segments lie below the curve

b. An overestimate because tangent segments lie above the curve

c. Exact value

d. Depends on the step size

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $\frac{dy}{dx} = 2x$ with $y(1) = 3$, what is the error in approximating $y(2)$ using Euler's method with $\Delta x = 0.5$?

a. 0.5

b. 1.0

c. 0

d. 0.25

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A population grows according to $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$. What is the carrying capacity of this population?

a. 500

b. 0.04

c. 250

d. $12,500$

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the logistic model $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$, at what population size is the growth rate $\frac{dP}{dt}$ maximized?

a. 250

b. 500

c. 125

d. 375

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is $\lim_{t \to \infty} P(t)$ for a logistic population with carrying capacity 800 if $P(0) = 50$?

a. 800

b. ∞

c. 400

d. 50

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the general form of the solution to the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{M})$?

a. $P(t) = \frac{M}{1 + Ae^{-kt}}$

b. $P(t) = Me^{kt}$

c. $P(t) = M - Ae^{-kt}$

d. $P(t) = \frac{M}{1 - Ae^{kt}}$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $P(t) = \frac{1000}{1 + 9e^{-0.2t}}$, what is the initial population $P(0)$?

a. 100

b. 1000

c. 90

d. 10

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the concavity of the logistic curve when $P(t) > \frac{M}{2}$?

a. Concave down ($P''(t) < 0$)

b. Concave up ($P''(t) > 0$)

c. Zero concavity

d. Undefined

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the autonomous differential equation $\frac{dy}{dx} = y(y-3)(y+2)$, what are all equilibrium solutions?

a. $y = 0, y = 3, y = -2$

b. $y = 3, y = -2$ only

c. $y = 0$ only

d. $x = 0, x = 3, x = -2$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For $\frac{dy}{dx} = 2y(1 - \frac{y}{4})$, which equilibrium solution is stable (an attractor)?

a. $y = 4$

b. $y = 0$

c. Both are stable

d. Neither is stable

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A population of bacteria is modeled by the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{L})$, where $P(t)$ is the population at time t in hours, k is the growth rate constant, and L is the carrying capacity. It is observed that $\frac{dP}{dt} = 200$ when $P = 100$, and the maximum growth rate occurs when $P = 500$. Find the carrying capacity L of this population.

a. 1000

b. 750

c. 1200

d. 500

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the logistic differential equation $\frac{dP}{dt} = kP(M - P)$, where $P(t)$ is the population at time t, k is the growth rate constant, and M is the carrying capacity. If $\frac{dP}{dt} = 0.05P(1000 - P)$, what is the carrying capacity of the population?

a. 1000

b. 0.05

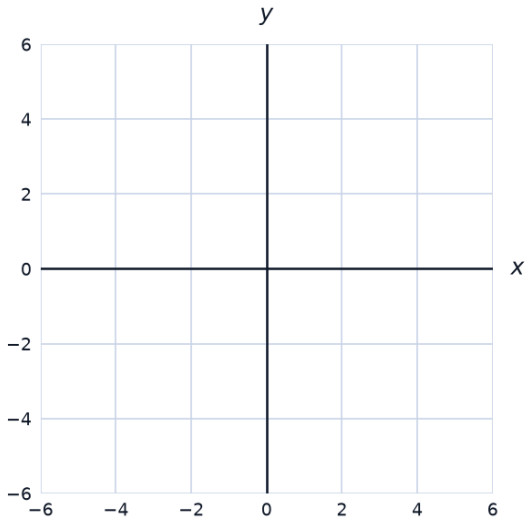
c. 50

d. 0.00005

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34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Logistic growth model is $\frac{dP}{dt} =$



a. kP

b. $kP(1 - P/M)$

c. $k(M - P)$

d. P^2

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A colony grows according to $\frac{dP}{dt} = 0.11P\left(1 - \frac{P}{1100}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=1100,\ P=550

b. M=1100,\ P=1100

c. M=550,\ P=1100

d. M=2200,\ P=550

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{2000}\right)$, find the instantaneous growth rate when $P = 1000$.

a. -25

b. 25.5

c. 25

d. 50

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider a region R in the xy-plane bounded by the graphs of $y = \frac{1}{2}x^2$ and $y = 2x$. A solid has its base in region R, and for each value of x in the interval $[0, 4]$, the cross-section perpendicular to the x-axis is an equilateral triangle with side length $s(x) = 2x - \frac{1}{2}x^2$. Write an integral expression that represents the volume of this solid. You do not need to evaluate the integral.

a. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right)^2 dx$

b. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right) dx$

c. $V = \int_0^4 \frac{1}{2}\left(2x - \frac{1}{2}x^2\right)^2 dx$

d. $V = \int_0^4 \frac{\sqrt{3}}{2}\left(2x - \frac{1}{2}x^2\right) dx$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 Taylor and Maclaurin Series

An AP Calculus BC FRQ models a population $P(t)$ satisfying the logistic differential equation $\frac{dP}{dt} = \frac{3}{600}P(600 - P)$. At what population size P is the population growing most rapidly?

a. $P = 300$

b. $P = 600$

c. $P = 150$

d. $P = 0$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Integration by Parts

Consider the function $f(x)$ whose derivative is given by $f'(x) = \frac{\sin(x)}{x^2 + 1}$. A related function $g(x)$ is defined by $g(x) = \int_0^x f(t)dt$. If $f(0) = k$, where k is a positive constant, find the value of $g'(0)$ and determine if $g(x)$ is increasing or decreasing at $x = 0$.

a. $g'(0) = k$, and $g(x)$ is increasing.

b. $g'(0) = 0$, and $g(x)$ is neither increasing nor decreasing.

c. $g'(0) = 1$, and $g(x)$ is increasing.

d. $g'(0) = k$, and $g(x)$ is decreasing.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

For $\frac{dy}{dx} = y,\ y(0) = 1$ (exact solution $y = e^x$), a single Euler step with $h = 0.25$ gives y_1. Compared with the exact value $y(0.25) = e^{0.25}$, Euler's approximation:

a. overestimates the true value

b. equals the true value exactly

c. cannot be compared

d. underestimates the true value

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Parametric Equations

On an AP Calculus BC Free-Response Question, a particle moves along a curve with velocity vector $\vec{v}(t) = \langle 2t, 3t^2 \rangle$. What is the speed of the particle at time $t = 1$?

a. $\sqrt{13}$

b. 5

c. $\sqrt{5}$

d. 13

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.02 Vector-Valued Functions

In a Taylor series BC FRQ, the alternating series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^3}$ is approximated by its first 2 terms S_2. According to the Alternating Series Error Bound, what is the maximum possible error $|S - S_2|$?

a. $\frac{1}{3^3} = \frac{1}{27}$

b. $\frac{1}{2^3} = \frac{1}{8}$

c. $\frac{1}{3^2} = \frac{1}{9}$

d. $\frac{1}{2 \cdot 3^3}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Polar Curves and Derivatives

For $\frac{dy}{dx} = y - x,\ y(0) = 1$, use Euler's method with step $h = 0.45$ to approximate y after two steps (at $x = 0.9$).

a. 1.45

b. 2.35

c. 1.9

d. 1.95

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.02 Area in Polar Coordinates

For $\frac{dP}{dt} = 0.4P\left(1 - \frac{P}{300}\right),\ P(0) = 60$, use one Euler step with $h = 1$ to approximate $P(1)$.

a. 158.4

b. 79.2

c. 60

d. 84

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

For $\frac{dy}{dx} = x + y,\ y(0) = 0$, use Euler's method with step $h = 0.1$ to approximate y after four steps (at $x = 0.4$).

a. 0.0641

b. -0.0359

c. 0.1641

d. 0.031

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